

THE GUTTER

Game Design Document · full team version

A 2.5D comic-book platformer about light, darkness and a hero who refused to die on page 3.



Concept still, part one (The Lit Pages): only what the Lamp lights is real and in colour.

Team only. This document contains the full story, including every twist and the ending. Do not share it with judges or players; the idea report is the public version.

How to use this document with Claude Code. Upload this PDF and treat it as the source of truth. Build one milestone at a time (section 12), in order, and check each milestone against its acceptance tests before moving on. All numbers marked as tuning values are starting points, not final. Engine: Unity 6 (latest LTS), Universal Render Pipeline (URP), C#, the new Input System, Cinemachine. A good first prompt: "Read this design document. Set up the Unity project structure from section 11, then build Milestone 1 only and stop."

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1. OVERVIEW AND DESIGN PILLARS

Pitch

A 2.5D comic-book platformer where light makes the world real, until the hero survives the page he was written to die on and the light starts hunting him.

At a glance

Item	Details
Genre	Puzzle platformer that turns into an action platformer (Hollow Knight-style combat)
Perspective	2.5D: full 3D scenes, side-on perspective camera, movement locked to a 2D plane (like New Super Mario Bros. and Paper Mario)
Platform	Windows PC; browser build on itch.io if the jam allows it
Engine	Unity 6 (latest LTS), URP, C#
Play time	About 25 to 40 minutes for the jam version
Competition themes	Comic, Light, Twist
Jam length	100 hours

Theme integration

Theme	How it appears
Comic	The game takes place inside a comic book. Rooms are panels, the narrator speaks in caption boxes, hits pop sound-effect text, and the final act pulls the camera out of the comic onto a real desk.
Light	Anything drawn is only solid and in colour while lit. Light changes meaning across the story: comfort, then threat, then acceptance.
Twist	The story is built around several plot twists, revealed through gameplay. The genre twists from puzzles to action. Gutter Mode lets the player literally rotate comic panels.

Design pillars

Every feature should support at least one of these. If a feature supports none, cut it.

- **Light decides what is real.** Every puzzle, hazard and fight comes from the three rules of light (section 5).
- **A comic you live inside.** Panels, captions, sound effects and ink are part of the world, not decoration on top of it.
- **The twist is played, not watched.** Reveals happen through mechanics: a fight you are meant to lose, a replay from the other side, control of the Lamp taken away and given back.

2. STORY (FULL, WITH SPOILERS)

Premise

Kip is the hero of a hand-drawn comic. A friendly narrator, the Writer, guides him through the story with caption boxes and the warm light of a desk lamp. Everything in the comic exists only where the Lamp shines. Kip is having a wonderful time, until page 3, where the script says he dies. He does not.

The story in the order the player experiences it

Beat	What happens	Purpose
1. Pages 1 to 2: The Lit Pages	Bright, funny comic world. The Writer jokes and helps. The player aims the Lamp to make platforms real. Scribbled shapes watch from the dark gutters between panels.	Teach the light rule. Build trust in the Writer and the Lamp.
2. Flash-forward: Page ??	Mid-level, a gust flips the pages forward. Kip is suddenly in the dark, ink-stained, holding a small ember, fighting monsters. He reaches the edge of a lit panel; a caption in the Writer's handwriting says "You shouldn't be here." The pages flip back.	Hook the player with a mystery and preview the action gameplay.
3. Page 3: the death page	The Shade slips into the lit world through a shadow cast by an object. Boss fight. The fight is designed to be lost. Kip's health hits zero and a "THE END" panel slams down. Silence. The player can still move: Kip crawls out through the panel border. The narration stutters: "...No. That isn't how this goes." The Lamp swings toward Kip and burns him.	The inciting incident. The light turns from friend to enemy.
4. The Margins	Kip flees into the gutters, the dark space where everything the Writer threw away ends up. Searchlight chases, combat, strange discarded characters. Eventually the player reaches the flash-forward moment, now with context.	Main action section. Mystery of why the light wants him gone.
5. Midpoint twist: the replay	Kip finds the crumpled draft of page 3. The player replays the fight as the Shade and discovers it was not attacking Kip; it was pulling him out of the panel before the death panel closed. The monster saved him.	Flip the player's understanding of the first act.
6. Second twist: the drafts	The monsters in the dark are earlier drafts of Kip, every version the Writer started and abandoned. Some are feral; some, like Draft #4, still talk.	Make the dark world tragic, not just scary.
7. Off the page: The Desk	Kip climbs out of the comic. The camera pulls back and the flat page becomes a full 3D desk at night. The Lamp was a real desk lamp; the gutters were the spine of the book.	The 2.5D reveal moment.
8. Final twist and ending	See below.	Emotional payoff.

The final twist

The Writer is not a villain. The comic is about someone the Writer lost (we suggest an older brother; the team can choose), and Kip is based on that person. Kip has to die on page 3 because that is what happened in real life. The Writer has redrafted the page again and again, unable to finish it. Every abandoned draft became a monster in the Margins.

The Shade is the final boss. It saved Kip because it wants the story never to end: if the last page is never drawn, nobody has to die. The Shade is denial.

Main ending: "The Last Page"

Kip beats the Shade on the desk and walks into the Lamp's light by choice. It no longer burns; it is warm again, like page 1. In a speech bubble, Kip tells the Writer it is okay. The Writer's hand draws the final panel but changes it: instead of dying in a fight, Kip waves goodbye and walks into the light. The camera pulls back to the comic's cover, "Issue #1, for [name]". Credits roll over the pages flipping, and now the dark pages are in colour too, because the Writer kept Kip's whole journey as part of the story.

Optional endings (only if time allows)

- **The blank ending:** Kip steals the pen and writes himself alive. The pages go blank white: a story with no ending, Kip alone forever. Gives the main ending more weight.
- **Lighter alternative:** the Writer and Kip co-write a new page 3 together, if the team prefers a less sad tone.

Why the light turns on Kip (the in-world logic)

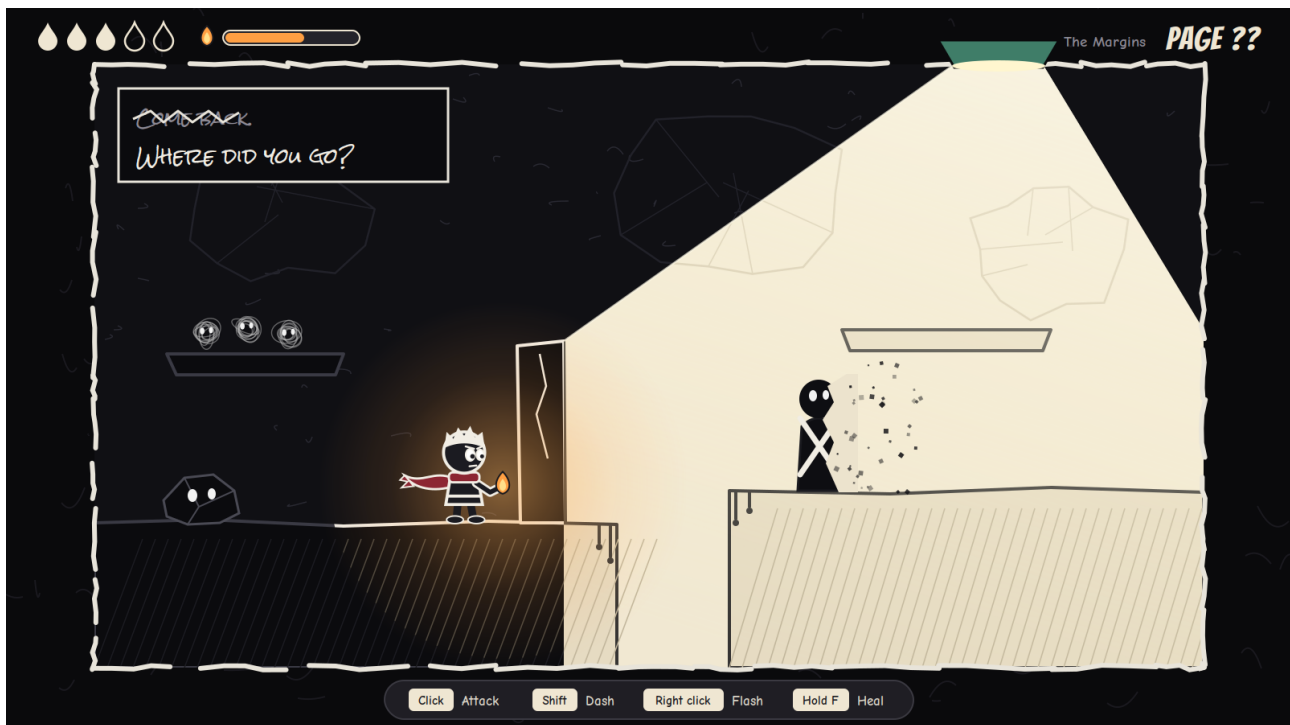
This is explained by the three rules of light (section 5). Before page 3, Kip is a written character, so the Lamp is safe for him. When he survives his death, the Writer crosses him out, which makes him a discarded thing, and the Lamp erases discarded things. In the ending, the Writer un-crosses him, which is why he can finally stand in the light.

3. CHARACTERS

Character	Role	Description and behaviour
Kip	Player character	A small comic hero with spiky black hair, big eyes, a blue tunic and a red scarf that is always blowing in the wind. Cheerful in part one, ink-stained and torn in part two. Carries an ember after page 3. Speaks only in short speech bubbles.
The Writer	Narrator, later antagonist, really a grieving person	Never shown directly until the ending, where only a hand is seen. Speaks through caption boxes: friendly and funny in part one (Comic Neue font), scratchy and frightened in part two (Rock Salt font, words crossed out).
The Lamp	Tool, then hazard	The Writer's desk lamp, green metal shade. In part one the player aims it. In part two the Writer uses it as a searchlight. In the finale the player gets control back.
The Shade	Apparent villain, secret rescuer, final boss	A tall shape of living ink with white slit eyes. Attacks Kip on page 3 (really pulling him out of the panel). Represents denial: wants the story never to end.
Draft #4	Guide NPC in the Margins	An earlier, unfinished version of Kip: half-sketched, no colour, missing an arm. Speaks plainly and sadly. Reveals the drafts twist.
Patch	Comic-relief NPC	A scruffy paper dog, a sidekick cut from draft 2 for being "too cute for the tone". Gives hints, overly enthusiastic.
Grimley	Comic-relief NPC	A villain replaced for not being scary enough. Sulks near bookmarks, tries hard to be menacing and fails.

4. THE THREE WORLDS

Part	Environment	Look and mood	Play style
1. The Lit Pages	Inside the comic, panel by panel: rooftops, streets, a town at night	Grayscale by default, colour wherever light falls. Warm, funny, safe.	Light-based puzzle platforming
2. The Margins	The dark gutters between panels, full of crumpled drafts and torn panel borders	Inverted: white ink lines on black. Lonely and tense. Inside the searchlight the world turns back to paper.	Exploration, stealth and combat
3. The Desk	Outside the comic: a real 3D desk at night, lamp, mug, pencils, the open book	Realistic lighting instead of ink lines, soft and cinematic.	Final boss combining all mechanics



Concept still, part two (The Margins): Kip hides in a pillar's shadow; inside the searchlight the page turns back to paper, and a Crossed-Out caught in the beam is being erased.



Concept still, part three (The Desk): Kip, tiny on the open comic, faces the Shade rising from the mug's shadow.

5. THE THREE RULES OF LIGHT

Every mechanic in the game follows from these three rules. Implement them first and keep them consistent; players learn the game by predicting them.

Rule	Meaning	Consequences
1. Light makes the drawn world real	Anything the Writer drew (platforms, bridges, doors, walls, ladders) is solid and in colour only while lit. Unlit, it is a gray, see-through ghost with no collision.	Puzzles are about getting light onto the right objects.
2. The Lamp erases anything crossed out	Monsters and discarded drafts are crossed-out things. The Lamp's light burns them away. Kip becomes crossed out after page 3.	Monsters hide in the dark. In part two, the Lamp hurts Kip. Monsters lured into the searchlight are erased.
3. Shadows count as darkness	A shadow inside a lit area follows dark rules.	A crate's shadow can erase a gate. Shadows are doorways for monsters in part one and hiding spots in part two.

Kip's own ember is not the Writer's light. It makes drawn things real (rule 1) but does not erase monsters (rule 2); it only stuns them with a flash.

6. MECHANICS IN DETAIL

6.1 Movement

Kip runs and jumps on a fixed 2D plane (X and Y); the Z position is locked. Use a Rigidbody with Freeze Position Z and freeze all rotations, or a custom kinematic controller.

- **Variable jump height:** holding jump goes higher; releasing early cuts upward velocity in half.
- **Coyote time:** Kip can still jump for a short moment after walking off a ledge.
- **Jump buffering:** a jump pressed just before landing still happens on landing.
- **Faster falling:** gravity is stronger while falling, so jumps feel snappy rather than floaty.
- **Ability unlocks:** part one has run, jump and a simple swing. Part two adds dash, downward strike and the ember.

6.2 Panels and camera

Each room is a comic panel with a thick border. When Kip crosses a panel edge, the camera snaps to the next panel. Panels can be different sizes (a tall panel for a climb, a wide panel for a chase). The strips between panels are the gutters.

- One Cinemachine virtual camera per panel; a trigger at each panel edge switches the active camera with a short blend.
- Neighbouring panels are visible at the screen edges, dimmed, so the player always feels inside a comic page.
- The final act uses one long camera move: pulling back from the page to reveal the 3D desk.

6.3 Light-reactive objects (rule 1)

- Every drawn object has a **LightReceiver** component that asks the light system "am I lit?" each physics tick.
- **Lit:** collider on, full colour. **Unlit:** collider off, grayscale, semi-transparent with a dashed outline.
- **Segments:** long platforms are built from short segments that check independently, so a beam can make part of a bridge solid.
- **Warning flicker:** when an object loses light it flickers briefly before turning ghostly, so the player is never dropped without warning.
- **Safe re-solidify:** when an object becomes lit while overlapping Kip, it waits until Kip is clear before turning its collider on.
- **Lit test:** a point is lit if it is inside a light's range and cone angle, and a raycast from the light to the point is not blocked by anything on the LightBlocker layer. Test 3 points per object (centre and both ends) and count it lit if any are lit.
- **Separate logic from looks:** never read the rendered lighting for gameplay. The gameplay check above decides; the visuals follow it.

6.4 The Lamp (part one and finale)

- The Lamp hangs at a fixed mount at the top of a panel. Its beam points toward the mouse cursor.
- The scroll wheel changes beam width: narrow reaches far, wide covers more area but reaches less far.
- Some panels have several mounts; clicking a mount moves the Lamp there. This enables multi-step puzzles.
- The beam erases monsters it touches (rule 2), so it is also the player's defence in part one.

6.5 Shadows (rule 3)

- Pushable crates (and other props on the LightBlocker layer) block light and cast shadows.
- A drawn object inside a shadow is unlit, so a crate pushed into the beam can make a locked gate passable.
- Every shadow is also a doorway: monsters in the gutters can reach into shadowed areas. Erasing an obstacle can let something in.

6.6 The Searchlight (part two)

- **Patrol:** sweeps a fixed pattern the player can learn and time.
- **Investigate:** moves to where Kip was last seen or where he used a flash.
- **Lock on:** follows Kip for a few seconds after spotting him.
- **Erase meter:** standing in the beam fills a meter (Kip's outline turns white from the edges inward). When full, Kip loses 1 health and is knocked out of the beam. In dedicated chase sequences, a full meter is instant death.
- **Warnings:** captions and a rising hum tell the player what the Lamp is about to do.
- **Using it:** the searchlight still makes drawn platforms solid (rule 1), so sometimes Kip must cross along the edge of the beam. Monsters lured into it are erased.

6.7 The Ember (part two onward)

- Kip's small personal light. It glows in a radius around him, reveals terrain line art and makes drawn objects solid within that radius.
- Runs on fuel (0 to 100). Hitting monsters adds fuel.
- **Flash:** spend fuel to blast the radius wide for a moment, stunning enemies and making distant platforms solid. It creates a noise that draws the searchlight.
- **Focus (heal):** hold to spend fuel and restore 1 health; Kip cannot move while channelling.
- The glow radius shrinks as fuel drops, which pushes the player to fight rather than hide forever.
- In the Margins, terrain is always solid but only visible near the ember; leaked pieces of the comic still need light to be solid.

6.8 Combat

- **Attack:** quick ink-blade slash, aimed left, right, up, or down (down only in the air).
- **Downward strike (pogo):** hitting something below bounces Kip up and resets his jump; used for traversal over spikes.
- **Dash:** short burst with a brief moment of invincibility; one air dash per jump.
- **Health:** 5 ink drops. After a hit, Kip flashes and is invincible for a short time.
- **Game feel:** hit-stop (a tiny freeze on hit), light screen shake and pop-up sound-effect text ("THWACK!", "SPLAT!").

6.9 Gutter Mode (stretch goal)

- At seams marked with ink drips, Kip steps into the gutter. Time freezes and the camera zooms out to show the whole page.
- The player clicks a panel and rotates it in 90° steps. Paths connect only where openings line up across the gutter.
- Lights inside a rotated panel rotate with it, so a beam can be turned to shine into the next panel.
- Locked panels have thick inked borders. The panel next to the seam Kip left from cannot rotate. Disabled near enemies.

6.10 Checkpoints, death and narration

- **Bookmarks** are checkpoints: resting heals, saves progress and resets enemies.
- **Ink stain:** in part two, dying leaves the ember's fuel behind as an ink stain. Touch it to recover the fuel; dying again first loses it.
- In part one, death simply restarts the current panel.
- **Captions:** event-driven caption boxes. Part one: jokes and gentle hints (after 3 failures on a puzzle). Part two: warnings that telegraph the searchlight ("Where did you go?" means searching, "There." means locked on).

7. ENEMIES AND BOSSES

Enemy	Where	Behaviour	Light interaction
Scribble	Parts 1 and 2	Small, fast, weak; attacks in swarms of 3 to 5.	Scatters from a flash; erased instantly by the Lamp.
Crumple	Part 2	A crumpled page that rolls at Kip; armoured.	Light (flash or searchlight) makes it unfold and become vulnerable for a few seconds.
Crossed-Out	Part 2	A tall discarded draft with a big X across its front that blocks hits.	Hit from behind, pogo off its head, or lure it into the searchlight.
Smudge	Part 2	Swims invisibly through shadows and lunges out.	Can only be hit while lit by the ember or a flash. Teaches how the Shade fights.

Bosses

Boss	Where	Phases and rules
The Shade (page 3)	End of part one	Phase 1: emerges from a shadow; the player burns it with the Lamp while Kip attacks. Phase 2: knocks props into the beam to make shadows and escape. Phase 3 (scripted): an unblockable attack takes Kip to 0 health, "THE END" slams down, music cuts, 3 seconds of silence, then holding any direction lets Kip crawl out through the border. Keep the whole fight to 60 to 90 seconds.
The Eraser (mid-boss)	Part two	A giant eraser that rubs out platforms under Kip, so the arena keeps shrinking. Its weak spot is exposed after it overreaches on a lunge.
The Shade (final)	The Desk	Hides in the shadows of desk objects (mug, pencil cup, books). It is erased by lamp light, but so is Kip. Phase 1: knock objects over to remove its hiding places. Phase 2: drive it into the light while staying out yourself. Phase 3: the Writer stops hunting Kip and the player gets mouse control of the Lamp again, aiming light and fighting at the same time.

8. LEVELS AND MISSIONS

Levels are comic pages, each made of several panels. The list below is the jam version. Each entry gives what the level teaches, so every new idea is introduced safely before it is tested.

Level	Part	Goal	Introduces or tests	Set piece
Page 1: Hello, Hero	1	Cross the rooftops to the bridge	Movement, captions, aiming the Lamp, lit platforms	Writer's friendly intro
Page 2: The Old Bridge	1	Cross the half-real bridge	Beam width, platform segments, pushing crates, shadow erases a gate	Flash-forward to the Margins mid-level
Page 3: The Death Page	1	Survive... or not	Shadows as monster doorways, multiple lamp mounts	Shade boss, "THE END", crawl out, first searchlight chase
M1: The Gutter's Edge	2	Escape the Lamp into the dark	Erase meter, hiding in shadows, ember radius	Chase sequence
M2: The Discard Pile	2	Find a way deeper	Combat, Scribbles and Crumples, flash, heal, bookmarks, ink stain	Meet Patch and Grimley
M3: The Sweep	2	Cross the searchlight fields	Searchlight stealth, Crossed-Outs, using the beam as a tool	Flash-forward scene now in context
M4: The Crumpled Draft	2	Find out what happened on page 3	Playable replay as the Shade	Midpoint twist; Draft #4 reveals the drafts
M5: Eraser's Hollow	2	Reach the edge of the page	Smudges, Gutter Mode puzzles (stretch)	Eraser mid-boss
D1: Off the Page	3	Climb out onto the desk	Traversal on real objects (ruler bridge, pencils, eraser)	Camera pull-out reveal
D2: The Last Page	3	Defeat the Shade	Everything: Lamp control plus combat	Final boss and ending

Level design guidelines

- Introduce each idea alone and safely, then combine it with one other idea, then test it under pressure.
- Every panel should have one clear "aha": the light trick or route that solves it.
- Place a bookmark before every boss and every hard searchlight section.
- Part one panels: 3 to 5 per page. Part two areas: 4 to 8 rooms each, with at least one bookmark.

9. CONTROLS

Action	Keyboard and mouse	Gamepad (optional)
Move	A / D	Left stick
Aim attacks up or down	W / S	Left stick
Jump	Space	A
Attack	Left click	X
Dash	Shift	Right trigger
Flash	Right click	Y
Focus (heal)	Hold F	Hold B
Interact / enter gutter	E	Right bumper
Aim the Lamp	Mouse position	Right stick
Beam width	Scroll wheel	Left / right bumper
Rotate panel (Gutter Mode)	Click a panel, then Q or E	Select with stick, rotate with bumpers
Pause	Esc	Start

In the finale the left hand moves Kip while the right hand aims the Lamp and attacks with the mouse, so the two control schemes from part one and part two combine naturally.

10. ART AND AUDIO DIRECTION

Visual style

- **Paper cutout 2.5D:** hand-drawn 2D sprites placed on flat planes at different depths in a 3D scene, lit by real 3D lights. This gives depth and real shadows without 3D character modelling.
- **Comic finish:** thick ink outlines, halftone dot shading, panel borders framing the screen.
- **Colour only in light** in part one; **inverted ink** in part two; **realistic soft lighting** on the desk.
- **Text as art:** captions, speech bubbles and sound effects are part of the scene.

Palette (starting values)

Use	Colour	Hex
Ink (outlines, text)	Near-black	#1A1A1F
Gutters and the dark	Deep charcoal	#17161B
Lamp light	Warm gold	#FFD65C
Ember	Orange	#FF9F43
Kip's scarf	Red	#E0313F
Kip's tunic	Blue	#2F6BD6
Caption boxes	Comic yellow	#FFE066
Paper	Warm off-white	#EFE7D4
Lamp shade	Green metal	#3F7D68

Typography

Font	Use	Licence
Bangers	Sound effects, titles, page numbers, boss names	Free (Google Fonts, OFL)
Comic Neue Bold	Writer captions in part one, speech bubbles, UI text	Free (Google Fonts, OFL)
Rock Salt	Writer captions in part two (scratchy, crossed-out words)	Free (Google Fonts, Apache)

User interface

- Top left: health as ink drops; ember meter with a flame icon (part two onward).
- Top right: area name and page number ("Page ??") in the Margins).
- Bottom: a small pill of control hints that changes per area, hidden after the player has used each control a few times.
- Boss fights: boss name and health bar at the top centre.

Audio

Part	Music	Key sounds
The Lit Pages	Light and playful: plucked strings, glockenspiel, soft percussion	Page flips, pen scratches, paper rustles
The Margins	Low drones, sparse percussion	Lamp hum rising in pitch before the searchlight moves, ink splats, scribble whispers
The Desk	A quiet piano theme building into the final fight	Real-world room tone, clock ticking, lamp switch click

11. TECHNICAL DESIGN FOR UNITY

Project setup

- Unity 6 (latest LTS) with the Universal Render Pipeline (URP) 3D template.
- Packages: Cinemachine, Input System, 2D Animation (for cutout sprite rigs), Shader Graph, TextMeshPro.
- Version control: Git with Git LFS for art and audio, through GitHub Desktop. Use Unity's standard .gitignore.
- Always tell AI tools the exact Unity version and that the project uses URP; code differs between render pipelines.

Folder structure

```
Assets/  
  _Project/  
    Art/           Sprites/ Materials/ Shaders/ Fonts/  
    Audio/         Music/ SFX/  
    Data/          Captions/ Enemies/ Tuning/      (ScriptableObjects)  
    Prefabs/       Player/ Lights/ Props/ Enemies/ UI/  
    Scenes/        Boot MainMenu Part1_Page1 ... Part3_Desk  
    Scripts/       Core/ Light/ Player/ Enemies/ Camera/ UI/ Narrative/
```

Layers

Layer	Used for
Player	Kip
Ground	Always-solid terrain
Drawn	Light-reactive objects (solid only when lit)
LightBlocker	Anything that casts gameplay shadows (crates, pillars, walls)
Enemy	Monsters and bosses
Trigger	Panel edges, bookmarks, caption triggers

Core scripts

Script	Responsibility
LightSource	Position, direction, range, cone angle; flags: isWritersLamp (erases crossed-out things), isEmber. Registers itself with LightSystem.
LightSystem	Singleton. IsLit(point, out source) checks all sources: range, angle, and a raycast against LightBlocker. Optional 20 Hz update for performance.
LightReceiver	On every drawn object or segment. Samples 3 points, asks LightSystem, toggles collider, sets the material _Lit value (0 to 1) with flicker and safe re-solidify.
LampController	Player-aimed lamp: follows the cursor, scroll changes width (wider angle, shorter range), mount switching.
SearchlightAI	State machine: Patrol, Investigate, LockOn. Listens for flash "noise" events.
EraseMeter	On Kip. Fills while inside a Writer's-lamp beam (after page 3), drains outside; on full: damage and knockback, or death in chase scenes.
PlayerController	Movement with coyote time, jump buffer, variable jump, dash. Z locked.
PlayerCombat	Directional attack, pogo bounce, hit-stop, refuels the ember on hit.
Ember	Fuel, radius from fuel, Flash and Focus abilities; it is also a LightSource with isEmber set.
Health	Shared by player and enemies: hit points, invincibility time, death events.
CrossedOut	Marker on monsters (and on Kip after page 3): if lit by a Writer's lamp, take erase damage.
PanelCamera	Switches Cinemachine cameras at panel edges.
GutterMode	Freeze time, page view, select and rotate panels, validate locked panels.
Bookmark / InkStain	Checkpoint save and reset; dropped fuel pickup.
CaptionSystem	Plays caption ScriptableObjects on events, with priority and cooldown.
GameState	Tracks part, page and story flags (e.g. kiplsCrossedOut), handles scene loading.

The lit check (pseudocode)

```
bool IsLit(Vector3 p, out LightSource by):
    foreach src in sources:
        v = p - src.position
        if v.magnitude > src.range: continue
        if Vector3.Angle(src.forward, v) > src.angle / 2: continue
        if Physics.Raycast(src.position, v.normalized, v.magnitude, LightBlockerMask): continue
        by = src; return true
    by = null; return false
```

Visual effects

- **Grayscale-to-colour:** one Shader Graph for drawn objects with a `_Lit` property (0 to 1) set by `LightReceiver`. At 0 it desaturates, lowers alpha and shows a dashed outline; at 1 it shows full colour. Visuals always match gameplay.
- **Comic look:** halftone dots in shadowed areas and an ink outline (inverted-hull or edge detection) in Shader Graph.
- **Margins look:** a "negative ink" material variant (black fill, white lines). A global shader property for the ember position and radius brightens line art near Kip. Inside the searchlight, switch back to the paper look using the same lit value.
- **Light visuals:** URP spot lights for the Lamp, plus a soft additive cone mesh for the visible beam and dust particles.

12. PRODUCTION PLAN

Milestones and acceptance tests

Build in this order. A milestone is done only when every test in its row passes.

Milestone	Hours	Build	Done when
M1: Light test	0 to 10	Project setup, Kip movement, one LightSource, LightReceiver platforms, one crate	Aiming the beam makes platform segments solid and coloured; unlit ones are ghostly and fall-through; the crate's shadow makes a gate passable.
M2: Core part one	10 to 35	Lamp control, panels and camera, shadows, captions, Scribbles in part one, Page 1 and 2 graybox	Pages 1 and 2 are playable start to finish in graybox; the Lamp erases Scribbles.
M3: Core part two	35 to 60	Ember, searchlight, erase meter, combat, dash, pogo, 3 enemy types, bookmarks, ink stain	One Margins area is playable: fight, refuel, flash, heal, hide from the searchlight, die and recover the stain.
M4: Content	60 to 80	All levels in the table in section 8, page 3 boss, desk scene and final boss, art integrated	The game is playable from start to ending. Feature freeze at hour 80.
M5: Polish	80 to 92	Audio, juice, menus, caption writing, balancing	A new player finishes without help; no placeholder art in key scenes.
M6: Ship	92 to 100	Bug fixing, builds, submission page	Windows build (and web build if allowed) tested on another computer; submitted.

Art (hours 5 to 85) and audio (hours 30 to 90) run in parallel with programming.

Team roles

Role	Responsibilities
Gameplay programmer	Player movement, combat, enemies, Lamp, ember, searchlight
Systems programmer and level designer	Light system, shadows, panel camera, Gutter Mode, checkpoints, building levels
Artist	Character and prop sprites, animation, comic shaders, the three environment looks
Audio, narrative and UI	Music, sound effects, caption writing, menus, submission page

If we run late, cut in this order

- 1 Gutter Mode (the plot twist still carries the Twist theme)
- 2 The Eraser mid-boss
- 3 The Smudge enemy
- 4 The flash-forward scene
- 5 Part two shrinks to a single area
- 6 The final fight is cut down to one phase

Never cut: the light rule, the page 3 "THE END" moment, and the camera pulling out to reveal the desk.

APPENDIX: TUNING VALUES

Starting points, in Unity units (Kip is about 1.6 units tall) and seconds. Put these in ScriptableObjects so they can be tuned without code changes.

System	Value	Start
Movement	Run speed	7 u/s
Movement	Jump height (full hold)	3 u
Movement	Early release cut	50% of upward speed
Movement	Fall gravity multiplier	2x
Movement	Coyote time / jump buffer	0.1 s / 0.1 s
Light	Ghost warning flicker	0.3 s
Light	Lit check rate	Every physics step (or 20 Hz)
Light	Platform segment length	1 u
Lamp	Narrow beam	20° angle, 14 u range
Lamp	Wide beam	60° angle, 7 u range
Searchlight	Patrol sweep speed	25° per second
Searchlight	Erase meter fill / drain	0.75 s / 1.5 s
Searchlight	Lock-on / investigate time	3 s / 4 s
Ember	Fuel / fuel per hit	100 / +10
Ember	Glow radius full / empty	3.5 u / 2 u
Ember	Flash	25 fuel, 9 u radius for 2 s, stun 1.5 s
Ember	Focus	33 fuel, 1 s channel, +1 health
Combat	Attack cycle / range	0.35 s / 1.5 u
Combat	Dash	0.2 s, 5 u, 0.5 s cooldown, 0.2 s invincible
Combat	Health / hit invincibility	5 / 1 s
Combat	Hit-stop	0.05 s
Combat	Pogo bounce speed	12 u/s
Enemies	Scribble / Crumple / Crossed-Out / Smudge health	2 / 4 / 5 / 3
Enemies	Crumple unfolded window	3 s
Bosses	Eraser / final Shade health	25 / 40 (3 phases)
Gutter Mode	Disabled within	6 u of an enemy